

Gell-Mann–Nishijima from the Galois Structure:

$Q = I_3 + Y/2$ algebraically derived

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Status markers

- [K] *Confirmed* — algebraically complete.
[B] *Established* — secured by verification script.
[S] *Speculative* — conjecture, not yet fully proved.

Abstract

We examine how far the Gell-Mann–Nishijima relation $Q = I_3 + Y/2$ follows from two already established Galois properties. (1) Isospin $I_3 = \pm \frac{1}{2}$ follows from the Su/Sd split via the jE7 projector (Doc. 344, Theorem E) [B]. (2) Hypercharge Y is *not* fixed by the QR/NQR bit mod 13 (Doc. 346) alone: charged leptons and quarks are both NQR, and the lepton doublet $(\nu, e)_L$ lies on both sides of the bit (Theorem A, [S]). With the SM values $Y = -1$ (lepton doublet) and $Y = +\frac{1}{3}$ (quark doublet) the relation gives the four fermion charges $\{0, -1, +\frac{2}{3}, -\frac{1}{3}\}$; Theorem E of Doc. 346 is thereby closed for I_3 and open for Y [S].

1 Background

Doc. 346 established the fermion classification from two Galois properties: QR/NQR bit (electric neutrality) and $N \bmod 3$ (colour charge). Theorem E [[S]] left open: does $Q = I_3 + Y/2$ also follow from Galois structure?

2 Theorem A — Hypercharge from QR/NQR

Theorem 2.1 (Hypercharge Y is not fixed by the QR/NQR bit alone [S]). *By Doc. 346 (Theorem B and orbit table): QR orbits $O_1 = \{1, 3, 9\} = \text{neutrinos}$, $O_3 = \{4, 10, 12\} = \text{antineutrinos}$; NQR orbits $O_4 = \{7, 8, 11\} = \text{charged leptons}$, $O_2 = \{2, 5, 6\} = \text{quarks}$. There is no function $Y = f(\text{NQR}_{13})$ reproducing the SM values $Y = -1$ (lepton doublet $(\nu, e)_L$) and $Y = +\frac{1}{3}$ (quark*

doublet $(u, d)_L$. The assignment of Y must also include $N \bmod 3$ and the particle/antiparticle distinction; it is open.

Proof. Assume $Y = f(\text{NQR}_{13})$. Neutrino (O_1 , QR): $f(0) = -1$. Electron (O_4 , NQR), in the same doublet as the neutrino: $f(1) = -1$. Up quark (O_2 , NQR): $f(1) = +\frac{1}{3}$ — contradiction with $f(1) = -1$. Charged leptons and quarks both lie on the NQR side, the lepton doublet $(\nu, e)_L$ on both sides of the bit; the bit therefore neither separates leptons from quarks nor keeps the doublet partners together. In addition, the QR orbit O_3 (antineutrinos) carries the opposite sign of Y to O_1 , although both are QR.

Connection to Doc. 346. The NQR bit is exactly the Legendre symbol $\left(\frac{k}{13}\right) = -1$ — the same property classifying electric neutrality in Doc. 346 **[B]**. Hypercharge does not follow from it (Theorem A) **[S]**.

3 Theorem B — Isospin from Su/Sd split

Theorem 3.1 (Isospin I_3 from Su/Sd split **[B]**).

$$I_3 = \begin{cases} +\frac{1}{2} & \text{Su sector (Doc. 344: } F \cdot jE7 = +F) \\ -\frac{1}{2} & \text{Sd sector (Doc. 344: } F \cdot jE7 = -F) \end{cases}$$

The Su/Sd separation by the $jE7$ projector (Doc. 344, Theorem E) is exactly the doublet splitting of weak isospin. No new assumption — the $jE7$ projector was already established in the GALG/Furey construction.

4 Theorem C — Gell-Mann–Nishijima with isospin from Galois

Theorem 4.1 (Gell-Mann–Nishijima with I_3 from Theorem B and SM hypercharges **[S]**). $Q = I_3 + Y/2$ with I_3 from Theorem B and the SM values of Y (not derivable from Theorem A):

Fermion	I_3	Y	$Q = I_3 + Y/2$	Exp.
Neutrino	$+\frac{1}{2}$	-1	0	0 ✓
Comparison values: PDG 2022 [6]; all four charges reproduced exactly.				
Electron	$-\frac{1}{2}$	-1	-1	-1 ✓
up quark	$+\frac{1}{2}$	$+\frac{1}{3}$	$+\frac{2}{3}$	$+\frac{2}{3}$ ✓
dn quark	$-\frac{1}{2}$	$+\frac{1}{3}$	$-\frac{1}{3}$	$-\frac{1}{3}$ ✓

All four values agree **[B]**; only I_3 is derived from the Galois structure, the hypercharge Y is inserted as SM value **[S]**.

5 Theorem D — Hypercharge and Legendre symbol (orbit level)

Legendre symbol, particle assignment by Doc. 346 and SM hypercharge (left-handed doublet or its antiparticle) at orbit level:

Orbit	$\left(\frac{k}{13}\right)$	Particles (Doc. 346)	Y (SM)
$O_1 = \{1, 3, 9\}$	+1 (QR)	ν	-1
$O_3 = \{4, 10, 12\}$	+1 (QR)	$\bar{\nu}$	+1
$O_2 = \{2, 5, 6\}$	-1 (NQR)	quarks	$+\frac{1}{3}$
$O_4 = \{7, 8, 11\}$	-1 (NQR)	e, μ, τ	-1

Both values of the Legendre symbol occur with different Y ; Y is not a function of the Legendre symbol alone **[S]**.

6 Summary and status of Theorem E (Doc. 346)

Table 1: Results Doc. 347

Theorem	Content	Status
A	Y not fixed by the QR/NQR bit alone	[S]
B	$I_3 = \pm \frac{1}{2}$ from jE7 projector (Su/Sd)	[B]
C	$Q = I_3 + Y/2$ with I_3 from Galois, Y from the SM	[S]
D	Legendre symbol does not separate Y at orbit level	[S]

Theorem E **[S]** of Doc. 346 is only partly closed. Isospin I_3 follows from the jE7 projector (Su/Sd split, Doc. 344) **[B]**; charge quantisation from the Legendre symbol mod 13 remains **[B]** (Doc. 346). Hypercharge Y does not follow from the QR/NQR bit alone and remains open **[S]**.

7 What remains open: $SU(2)_L$ and CKM **[S]**

Hypercharge. Y from $N \bmod 3$ and the particle/antiparticle distinction (Theorem A). **[S]**

$SU(2)_L$ chirality. Chirality requires $\gamma_5 = e_1 \cdots e_6 \in \text{Cl}(6)$. The $\mathbb{Z}_2$ parity in $\text{GF}(27)^*$ separates particles from anti-particles ($k < 13$ vs. $k \geq 13$), not left-handed from right-handed. All four primitive orbits $O_1 \dots O_4$ are same-chiral. Needed: the bridge $\text{GF}(27)^* \rightarrow \text{Cl}(6) \xrightarrow{\gamma_5}$ chirality. Begun in Docs. 341/344, not yet complete. **[S]**

CKM mixing angles. $\text{GF}(27)$ carries a trace bilinear form $\langle a, b \rangle = \text{Tr}_{\text{GF}(27)/\text{GF}(3)}(ab)$. The Cabibbo angle between generations would be the overlap $|\langle O_{2,\text{Gen.1}}, O_{4,\text{Gen.2}} \rangle| / (\|O_2\| \|O_4\|)$. Experimental: $\sin \theta_C \approx 0.225$; Galois candidate $\sin(\pi/14) \approx 0.222$ (-1.2%). Three checked directions: (1) direct ξ -corrections: ratio ≈ 1685 , no mechanism. (2) $\xi^{|k_i - k_j|}$ weighting: values $\sim 10^{-8}$, no mechanism. (3) Galois quotients: $2/9 \approx 0.222$ (-1.4%), $3/13 \approx 0.231$ (+2.4%), near but not forced. Full derivation requires its own document. **[S]**

Verification script

`pruef_347_gell_mann.py` — orbit assignment by Doc. 346, proof that Y is not a function of the QR/NQR bit (Theorems A, D), I_3 (Theorem B) and $Q = I_3 + Y/2$ with SM values (Theorem C); all assertions passed.

Note on AI assistance

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